Las Vegas

May 31 – Jun 4, 2026

LABATO-2002 - Gain Real-Time Visibility into Asset Utilization with Cisco SNA and Splunk

Walk-In Labs

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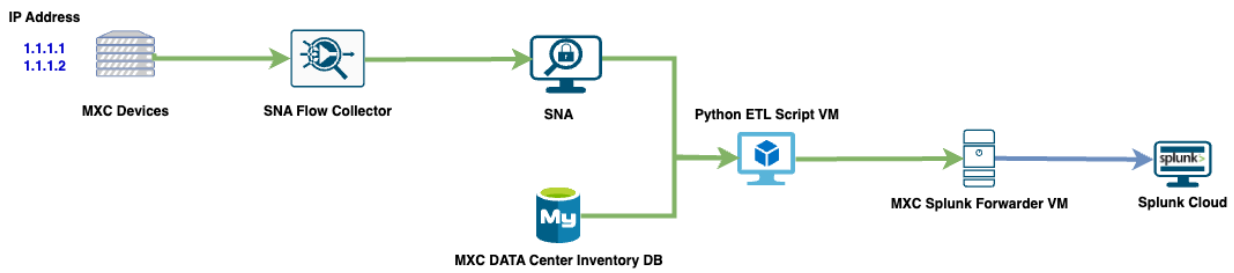
Overview

This lab provides step-by-step instructions on how to configure Cisco Secure Network Analytics (SNA) to monitor device-level flow counts and export network telemetry data to Splunk for centralized analysis and reporting. (Expand/Add)

Scenario: Unified Power Management Across Distributed Data Centers

In this lab, you are a network engineer responsible for managing a data center environment. Your primary objective is to monitor device access by configuring Cisco Secure Network Analytics (SNA) to track asset utilization using their IP addresses. Through this configuration, you will collect detailed telemetry data, including flow records, to analyze how frequently each device is accessed and understand network activity patterns.

Additionally, you will learn how to integrate and visualize SNA telemetry data within Splunk Cloud. This integration enables you to enrich network insights by leveraging Splunk's powerful analytics and dashboard capabilities, providing a unified view of device utilization and network behavior. By the end of this lab, you will be equipped to transform raw network flow data into actionable intelligence, helping optimize asset usage and improve operational efficiency.



***Ram has updated topology!!**

Learning Objectives

Upon completion of this lab, you will be able to:

- [Add line for checking device status?]
- Configure and monitor host groups effectively within Cisco Secure Network Analytics (SNA), formerly known as Stealthwatch, to organize and track device activity.
- Analyze access trends by evaluating the unique Flow count of assets, using flow search in SNA and visualizing data telemetry in Splunk.
- Develop and propose solutions to optimize inventory management based on device usage insights.

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Note

The datacenter entry gateway is pre-configured for NetFlow, streaming telemetry to the SNA Flow Collector for analysis. Additionally, the SNA virtual appliances has been pre-provisioned to support the tasks in this

Info

What is a Flow Collector? ** Change this, include definition of Netflow**

A Flow Collector is a core component of the Cisco Secure Network Analytics (SNA) architecture. Its primary function is to ingest, deduplicate, and store flow data (such as NetFlow) exported from network devices. By processing this telemetry, the Flow Collector provides the Stealthwatch Management Console (SMC) with the granular data necessary for network-wide visibility, behavioral analysis, and threat detection.

Why is Visibility into Device Access Important?

Maintaining visibility into how often devices are accessed within the data center is critical for several reasons:

- Security Posture: Tracking access patterns allows administrators to identify anomalous behavior, such as unauthorized lateral movement or potential data exfiltration attempts.
- Compliance: Many regulatory frameworks require organizations to maintain logs of who accessed which resources, for how long, and with what frequency.
- Troubleshooting: Detailed flow logs provide a historical audit trail, making it easier to diagnose connectivity issues or performance bottlenecks between endpoints.

Benefits of Monitoring for Capacity Planning

Beyond security, monitoring device access provides actionable insights for infrastructure management:

- Predictive Resource Allocation: By analyzing traffic trends, administrators can identify highly utilized servers or services. This enables proactive scaling decisions, such as adding hardware or increasing bandwidth, to prevent performance degradation before it occurs.
- Optimization of Assets: Monitoring reveals underutilized devices, allowing IT teams to consolidate workloads, reduce power consumption, and optimize the overall data center footprint.
- Trend Analysis: Establishing a baseline of "normal" traffic patterns helps in forecasting future growth requirements, ensuring that the data center infrastructure evolves in alignment with business demands.

Disclaimer

This lab was designed exclusively for educational and training purposes. The activities performed within this environment are conducted on controlled dashboards and do not reflect real-time changes to live production infrastructure.

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Getting Started

Access Requirements

a. Connectivity and Hardware
   - **Network Access:** Please ensure you have a stable internet connection for the duration of the session.
   - **Recommended Hardware:** A laptop or tablet is recommended to ensure full compatibility and optimal interaction with the dashboard interfaces.
   - **VPN Client:** Cisco Secure Client
   - **Terminal Emulator:** PuTTY
   - **Browser:** Google Chrome

b. SNA Virtual Appliances – Provided in this lab.
   - SNA Data Node (v7.6.0)
   - SNA Flow Collector (v7.6.0)
   - SNA Manager (v7.6.0)

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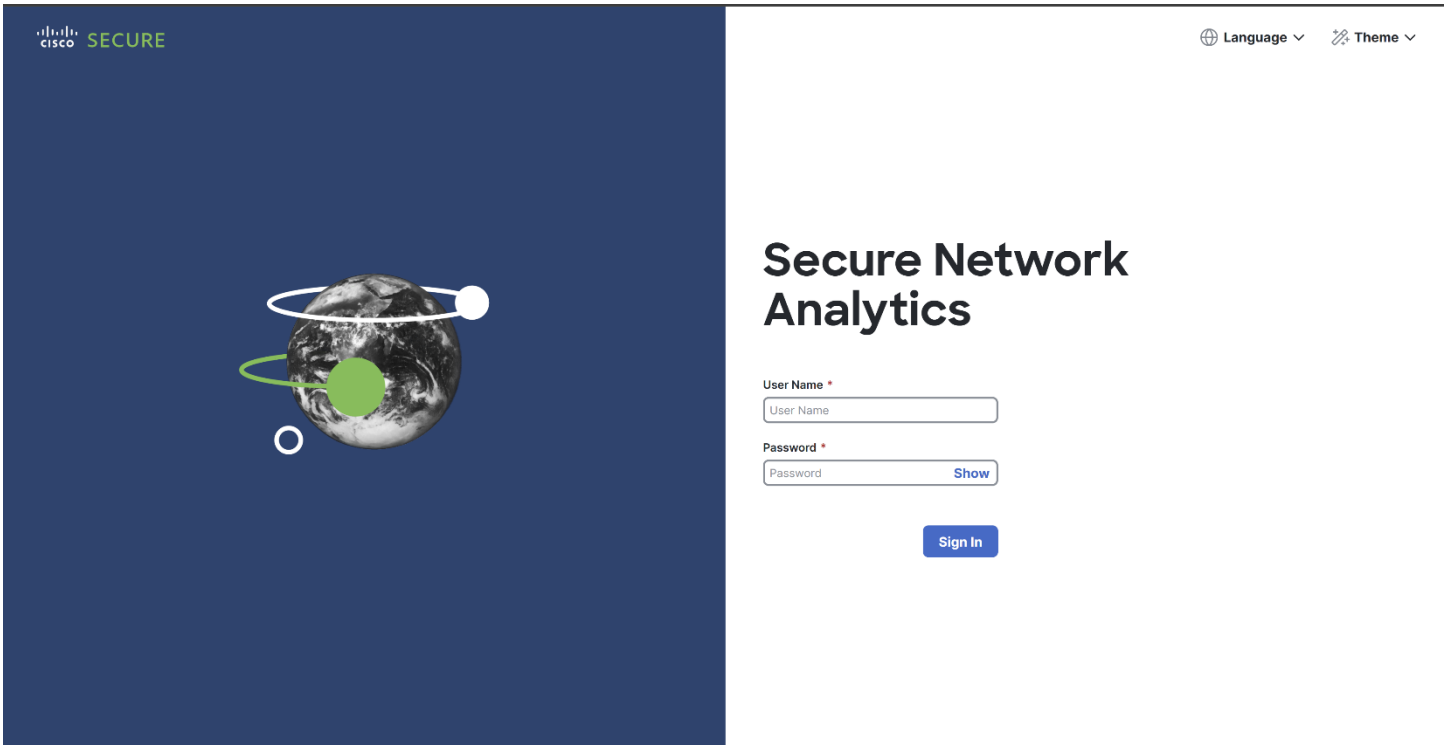
Task 1: SNA Appliance Setup

In this task, we will demonstrate how to navigate the SNA dashboard's *Central Management* tool and view the status and settings of the virtual appliances that comprise your data center's SNA environment. Each virtual appliance plays a vital role in this system, and it is crucial to verify they are correctly configured and operational. Through this process, you will be familiarized with the essential components of SNA and confirm that your deployment is ready to collect and monitor flow data.

Step 1: Accessing the SNA Dashboard

In Task 1-2, no admin access to configure these settings, later in task 4.

a. Enter your username and password, then click **Sign In** **[HIGHLIGHT ALL ITEMS IN PICTURE]**



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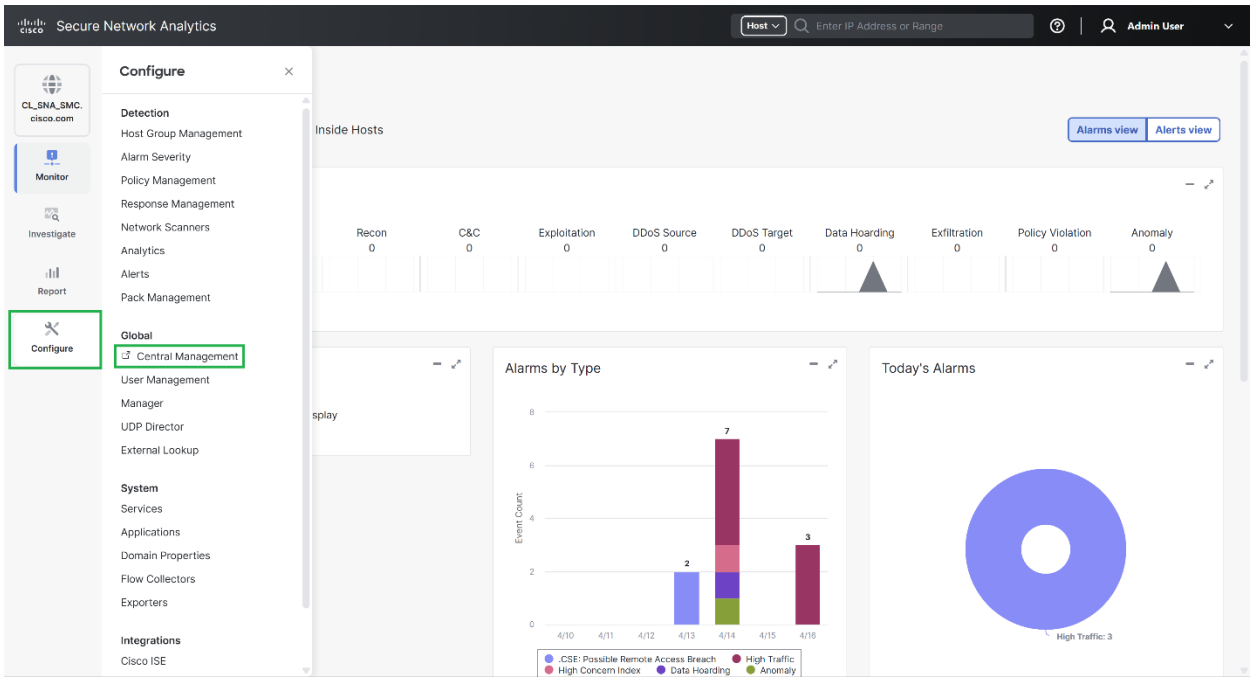
Step 2: Navigating to Central Management

Central Management is the centralized management console for your SNA virtual appliances. It is a tool that provides full control over each appliance and detailed insight into their operations. Using Central Management, you will be able to edit your appliance settings and view their analytics data.

a. In the sidebar, navigate to **Configure** > **Central Management**



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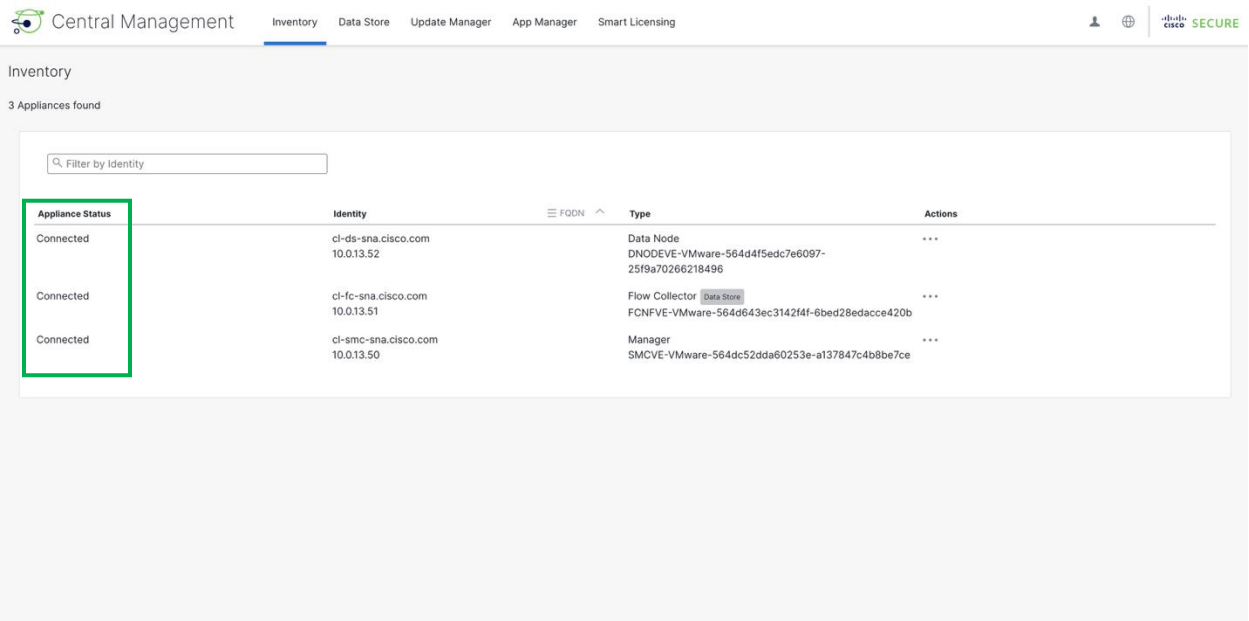
Step 3: Verifying Virtual Appliance Status

To ensure your SNA environment is operational, you will check the status of each virtual appliance in the *Inventory* tab. This tab is the default view for Central Management.

a. In the *Appliance Status* column, verify that each appliance is **Connected.** **[3a]**



[NOTE]The three core appliances of SNA architecture, which your environment mirrors, are as follows:

a. Secure Network Analytics Manager: Provides the graphical user interface for managing the SNA environment and analyzing captured data.
b. Data Node: Acts as the storage repository for captured data.
c. Flow Collector: Responsible for collecting and forwarding network telemetry to the Data Node.

Critical

When bringing an SNA environment online, it is essential that the appliances are deployed in the correct order. Failure to do so will prevent them from connecting to the SNA Manager, rendering the environment non-operational. The correct order of deployment is **SNA Manager > Data Node > Flow Collector**.

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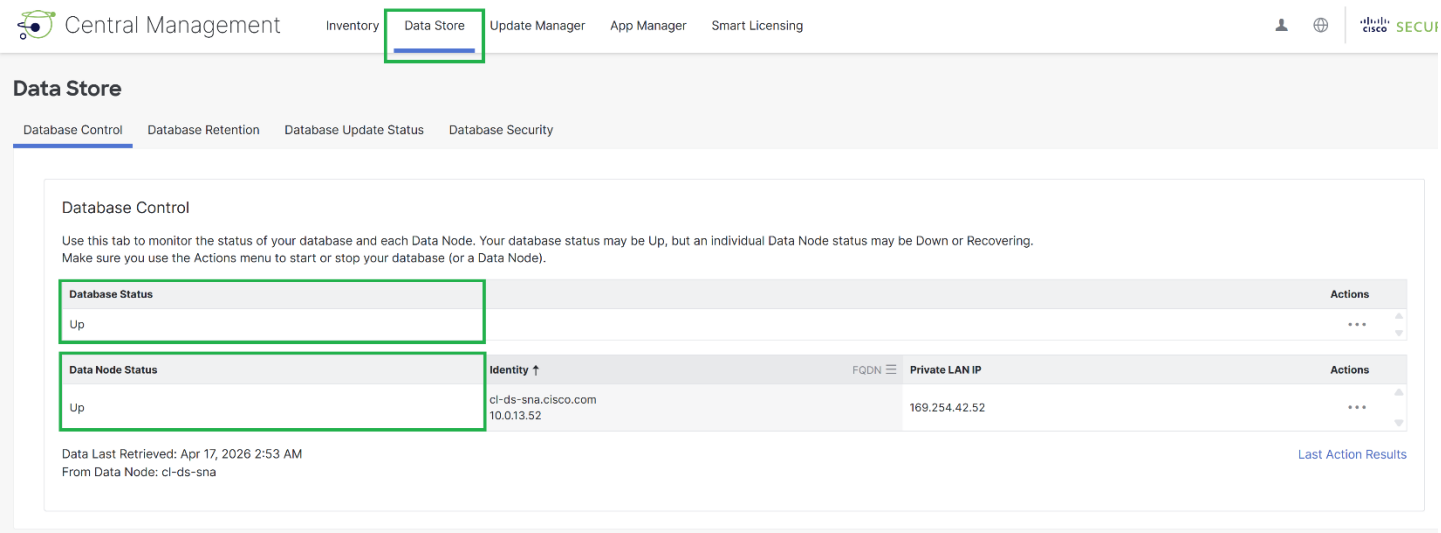
Step 3: Verifying Data Node Status

The Data Node is an appliance that stores, indexes, and retrieves the flow data captured by Flow Collectors. The data center generates a vast amount of network traffic, so a dedicated storage component is imperative to your SNA environment. While the information is physically stored on the Data Node, it resides logically in a database. For the database to be accessible, the Data Node must also be online and connected. You will check the Data Store page to view the operational status of both the Data Node and database.

1. In the banner, navigate to **Data Store**. The default view is the *Database Control* tab.
2. Under *Database Status*, verify it is **Up**.
3. Under *Data Node Status*, verify it is **Up**.



Note or include in Step 3 Summary?

[The Data Store is a central database, consisting of one or more Data Node[s], that stores, indexes, and retrieves the captured network telemetry. Your Data Store is made up of only one appliance, but the clustering of multiple Data Nodes provides high-availability and can improve both fault tolerance and query-response times.]

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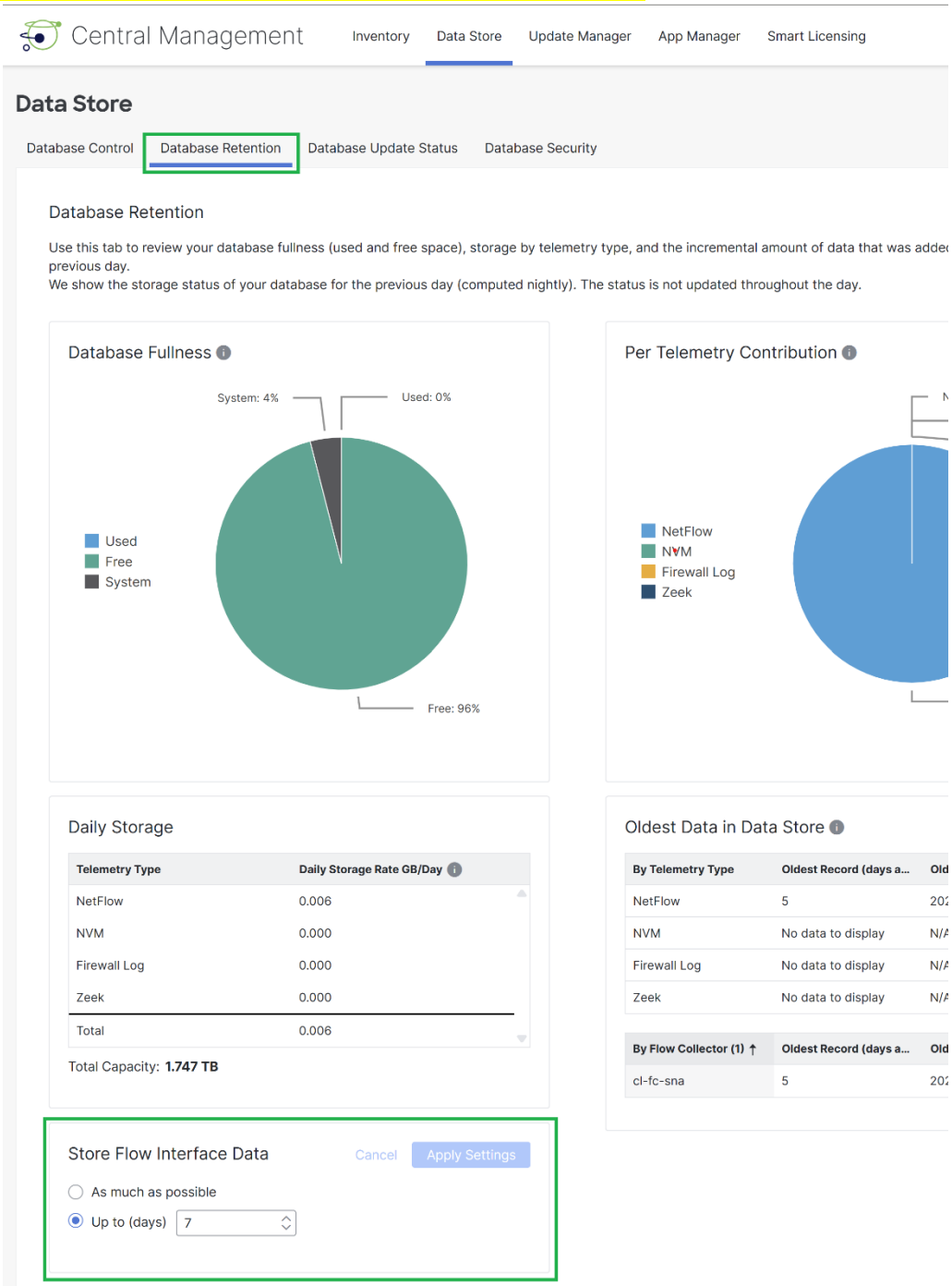
Step 5: Verifying Retention Period

One of the benefits that SNA offers is the ability to analyze historical data, but it also allows you to decide how long you will hold onto that data. By configuring the Store Flow Interface Data policy, a retention period can be set that meets your data center's needs. You plan to locally store your SNA's historical data locally in the Splunk index, so a seven-day retention policy will be adequate here.

a. Navigate to **Database Retention**.
b. Under *Store Flow Interface Data*, verify that **Up to (days)** is selected, and the value is **7.**

[Edit Picture Full Screen, but don't need to scroll to view bottom green box]



[Task Conclusion]

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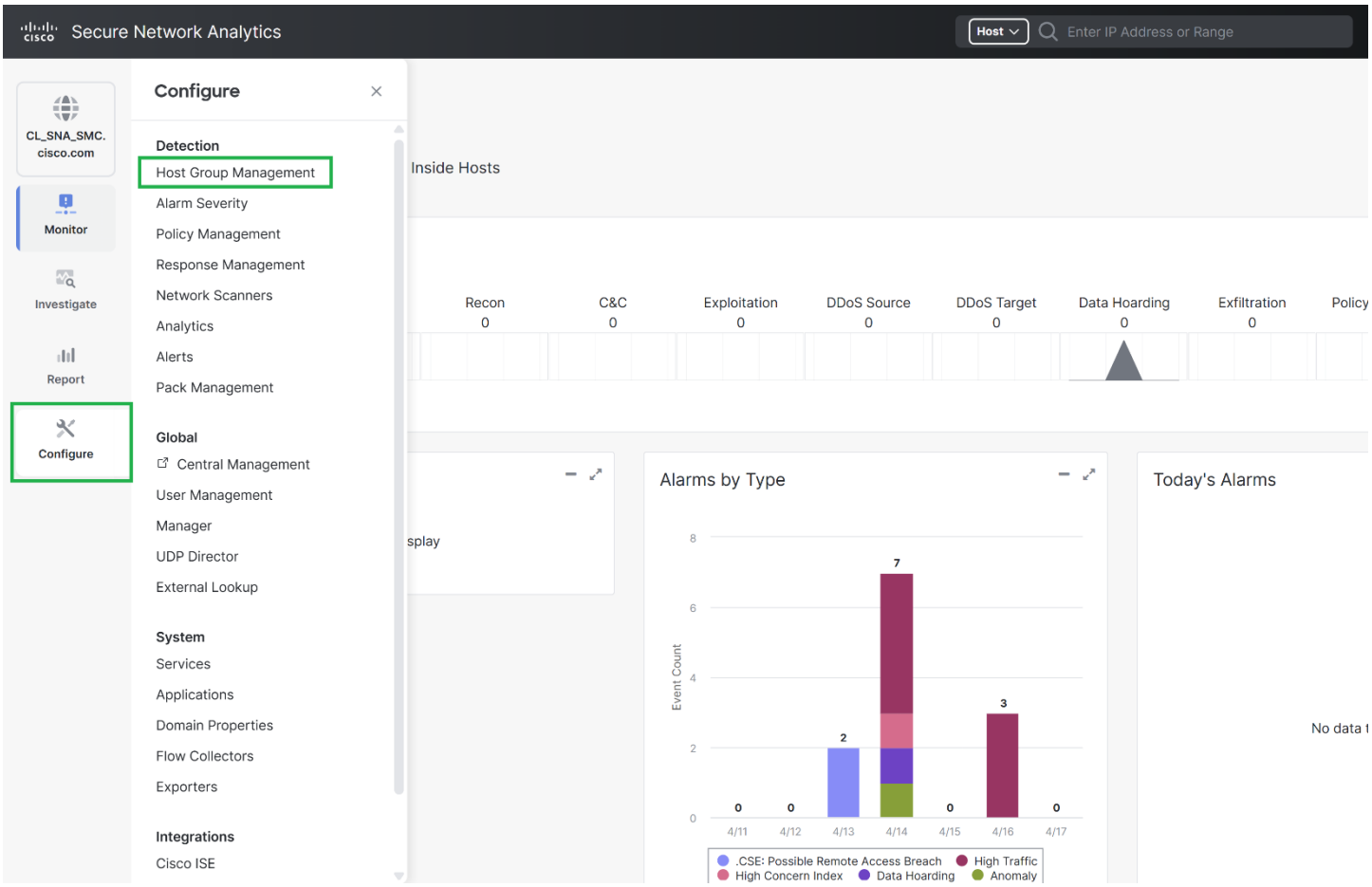
Task 2: Monitoring Asset Utilization – Host Group Management

Now that you have confirmed that the SNA appliances are connected and functional, they can be used to monitor asset utilization in your data center. The assets you want to track should be configured in the Host Group Management, by providing that subnet's information to the SNA dashboard, your appliances will begin capturing and storing that network's traffic data. For this task, we will demonstrate how to add a Host Group in the SNA and assign the asset's subnet to it.

Step 1: Navigating to Host Group Management

a. From SNA Manager, navigate to **Configure** > **Host Group Management**



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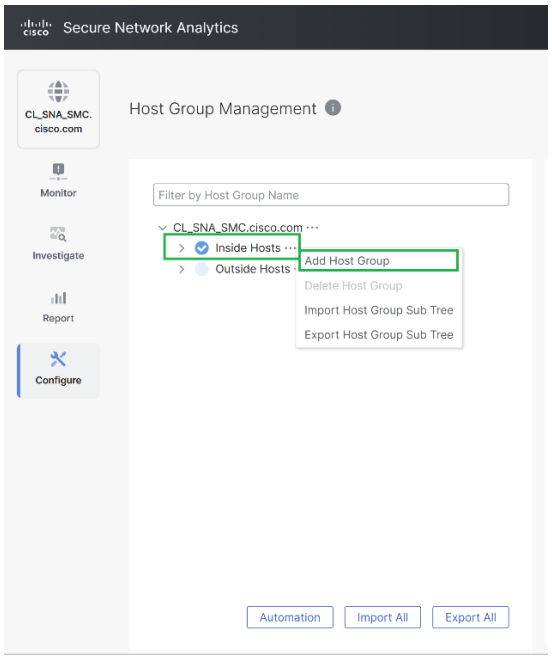
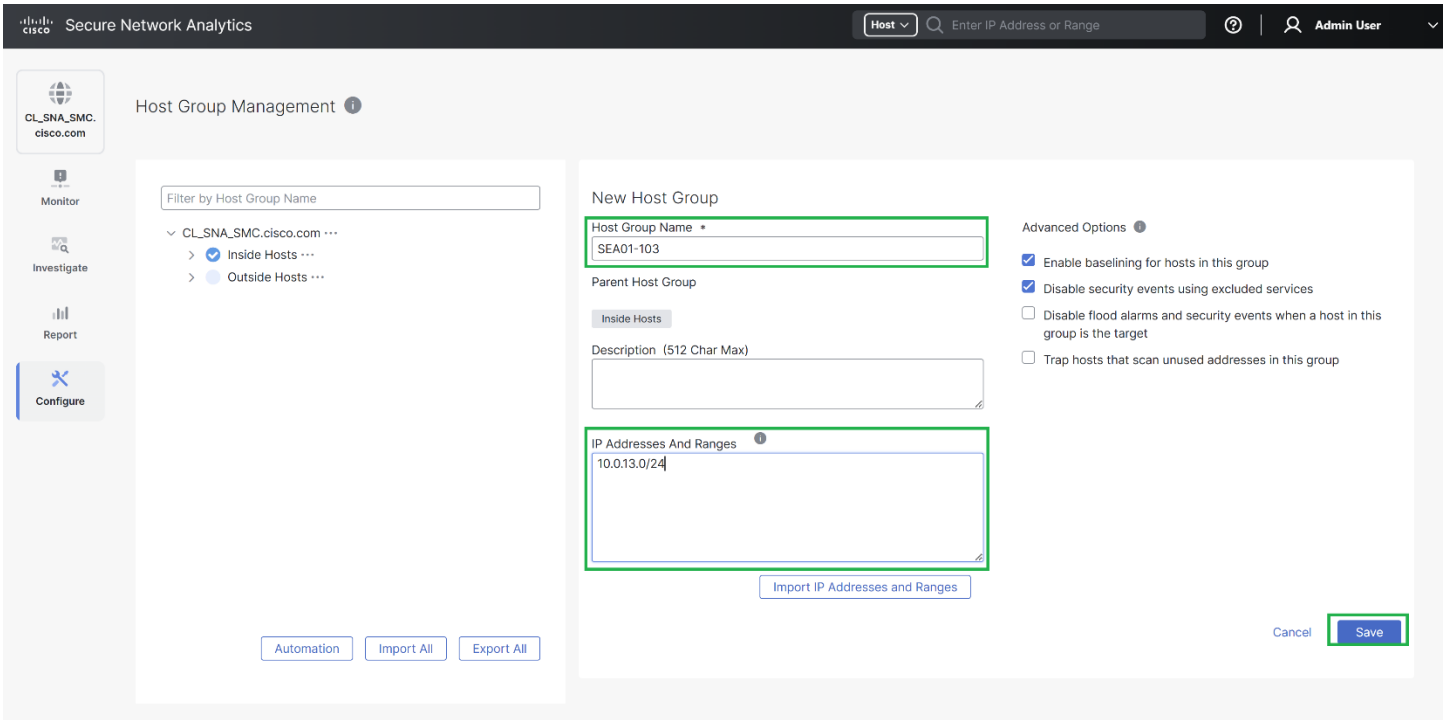
Step 2: Adding a Host Group - No admin access, just a demonstration [Note]

Within a single domain, Host Groups are added to one of two categories in the SNA: *Inside Hosts* for trusted internal assets, or *Outside Hosts* for untrusted external hosts. This segmentation allows for the application of detailed security policies and deeper insight into traffic patterns. The assets from the SEA01-103 data center are internal and trusted, so you will add them to *Inside Hosts*.

a. Next to *Inside Hosts,* select the **(...)** icon.
b. In the popup menu that appears, select **Add Host Group**.



c. Under *Host Group Name*, enter **SEA01-103**
d. Under *IP Addresses and Ranges*, enter **10.0.13.0/24**
e. Select the **Save** button to add the Host Group



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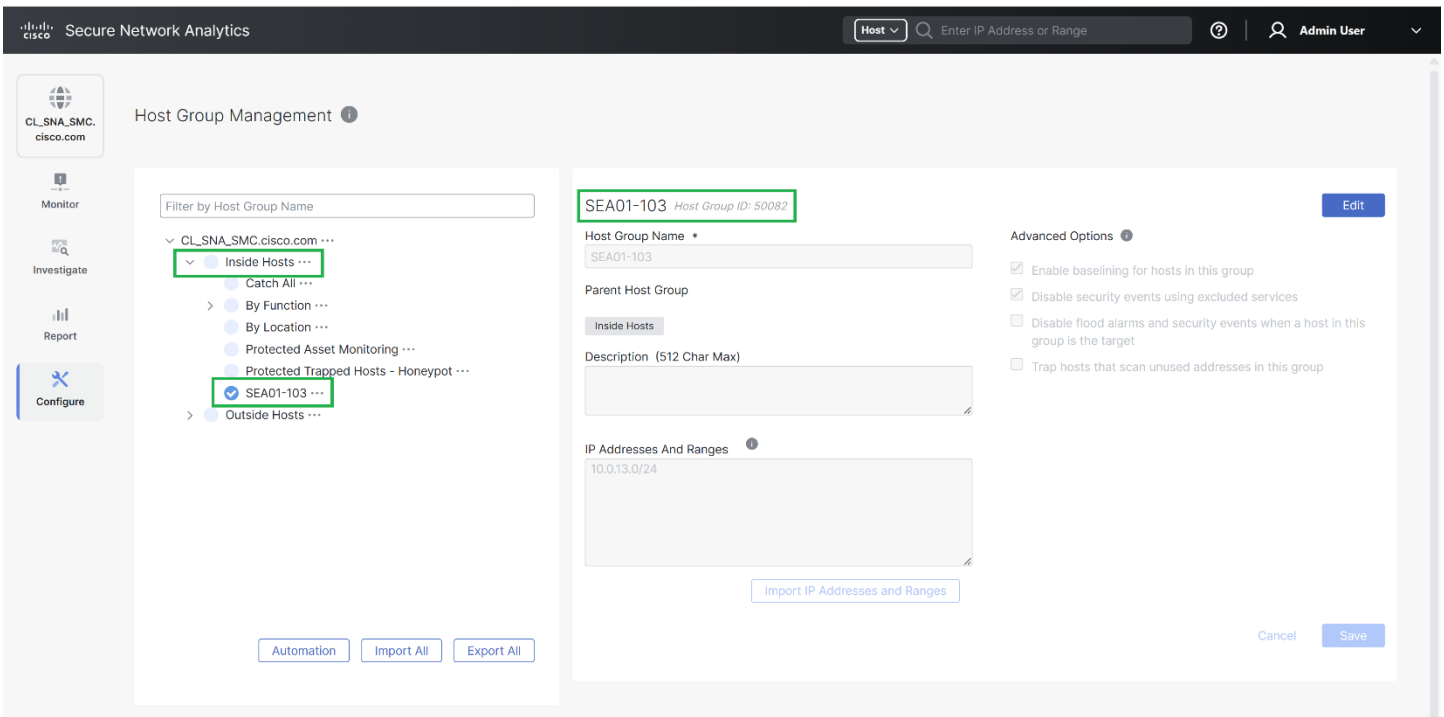
Step 3: Verifying the New Host Group is Added

After adding your Host Group, it is important to check that it was created successfully.

f. Select the dropdown arrow **(>)** next to *Inside Hosts* to expand the list.
g. Select **SEA01-103** to display the Host Group information.

[Edit Picture, remove box over Host Group ID]



[Task 2 Conclusion]

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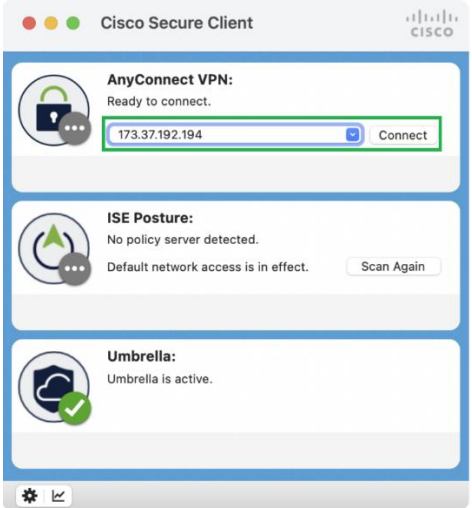
Task 3: Simulating Asset Utilization Using NetFlow

[Task 3 Summary/Expectation/Purpose] To verify the SNA environment is capturing NetFlow as intended, you will access SEA01-103 data center assets via SSH and HTTPS to generate network traffic data. By simulating network traffic with multiple access protocols, you will

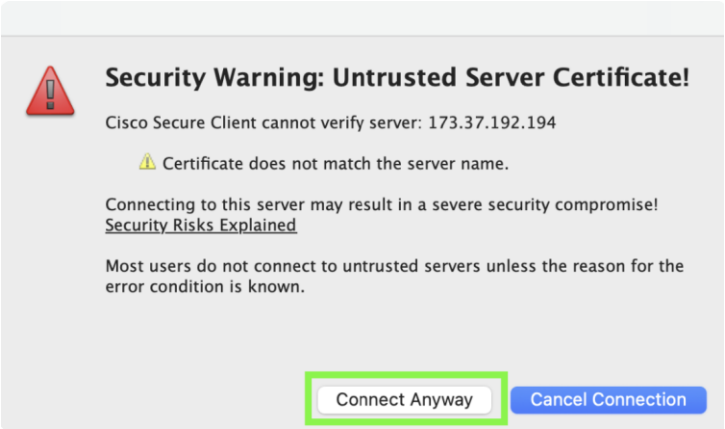
Step 1: Connecting to the VPN

For this task, you will need to connect to the lab environment through our VPN.

a. Open **Cisco Secure Client**, enter the VPN address: **173.37.192.194** and select **Connect**.



b. If prompted with a security warning, select **Connect Anyway** to proceed.



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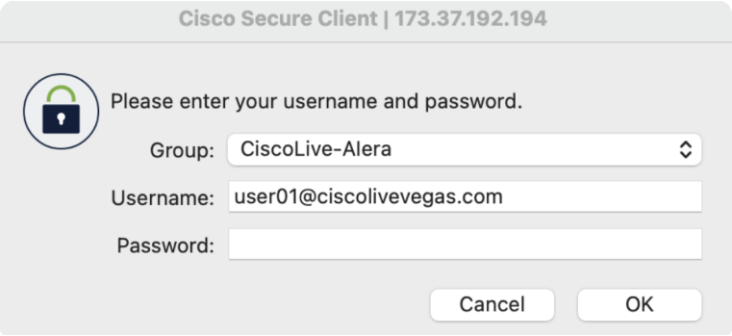
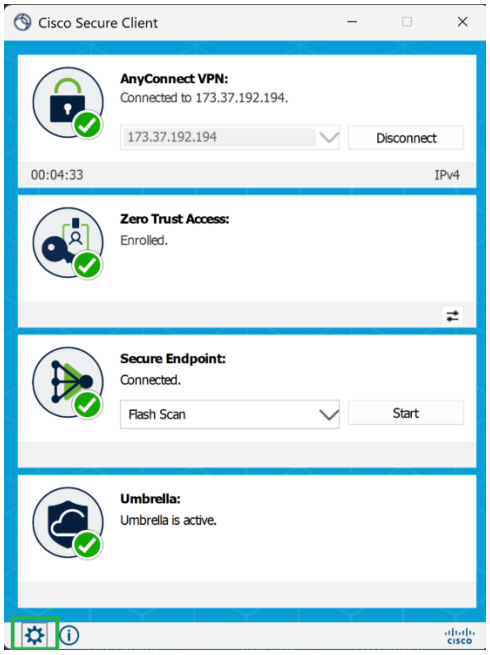
c. Enter the credentials listed below and select **OK** to connect

Username	**user01@ciscolivevegas.com**
Password	**eaHZP_a8zphk*ty**



d. After successfully connecting, reopen **Cisco Secure Client** and select: [Settings Icon]



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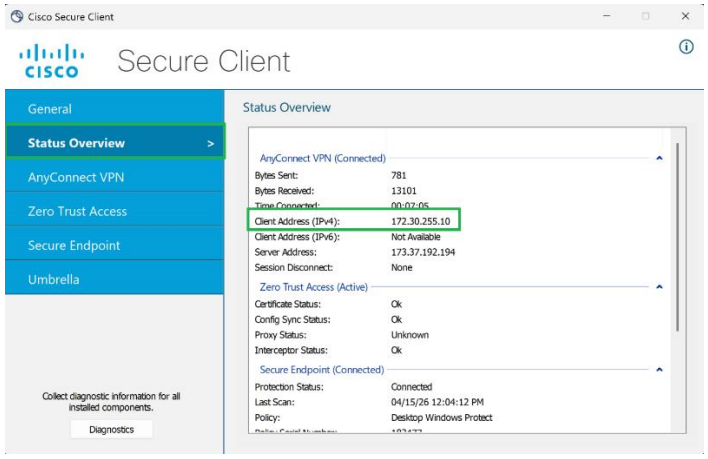
e. Navigate to **Status Overview**
f. Note down the IP Address next to *Client Address (IPv4)*. You will need to reference it later in the next Task. In the example, the IP Address is **172.30.255.10**.

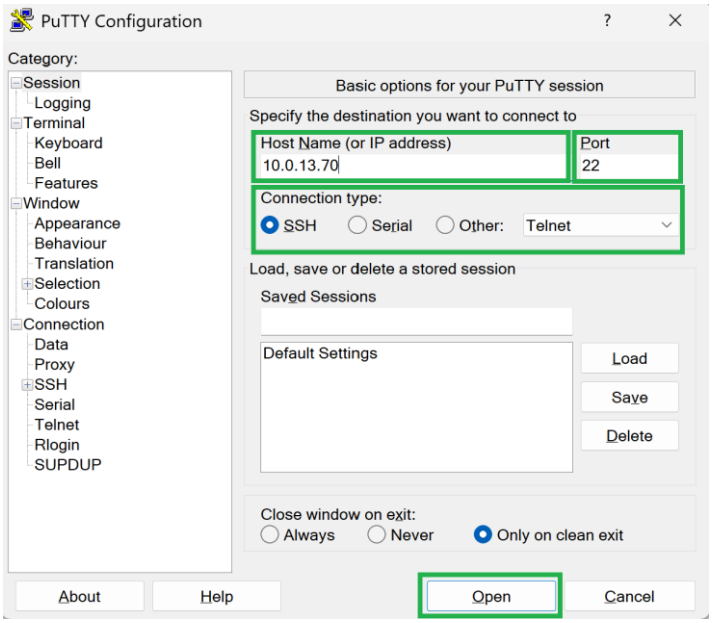


Step 2: Accessing the Device: ISR4K-CL via SSH

The first asset in the SEA01-103 data center that you will access is a Cisco ISR4451 router. Using PuTTY, you will connect to the command-line interface via SSH. The access information is provided below.

Host Name	Mgmt IP	Access Method	Username	Password
ISR4K-CL	**10.0.13.70**	**SSH**	**aiera-user**	**Ciscolive!135**

a. Open **PuTTY**, then enter **10.0.13.70** in the *Host Name (or IP Address)* field.
b. Verify that *Port* is **22** and *Connection type:* is **SSH**, then select **Open**.



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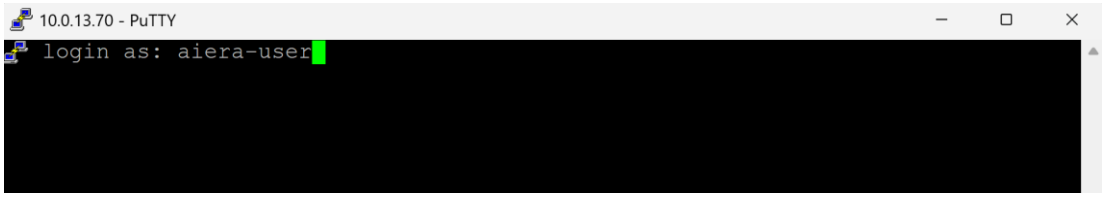
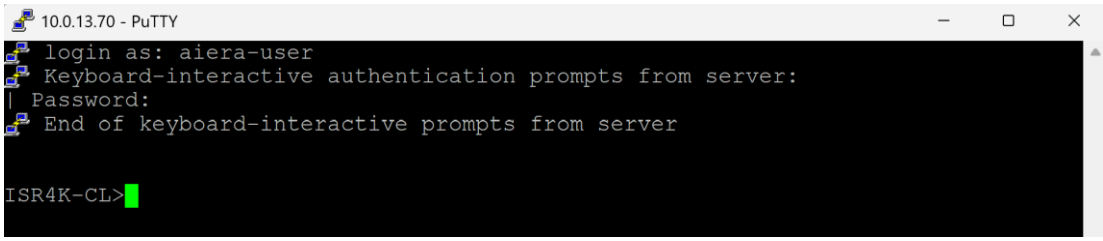
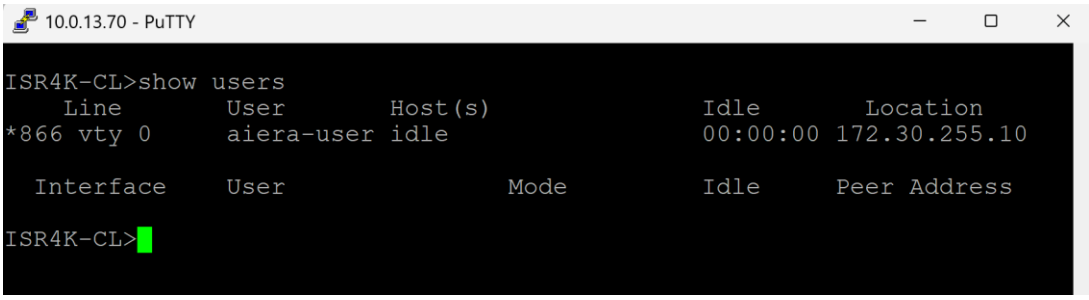
c. The asset terminal will open. At the login prompt, enter the username: **aiera-user** and press *Enter*.



d. At the password prompt that follows, enter the password: **Ciscolive!135** and press *Enter*. When entered correctly, you will see the enable prompt for *ISR4K-CL*.



e. At the enable prompt, enter the command: **show ip interface brief** and press *Enter* to simulate additional network traffic to this asset. [Edit Picture]



f. Close the terminal window to exit the session.

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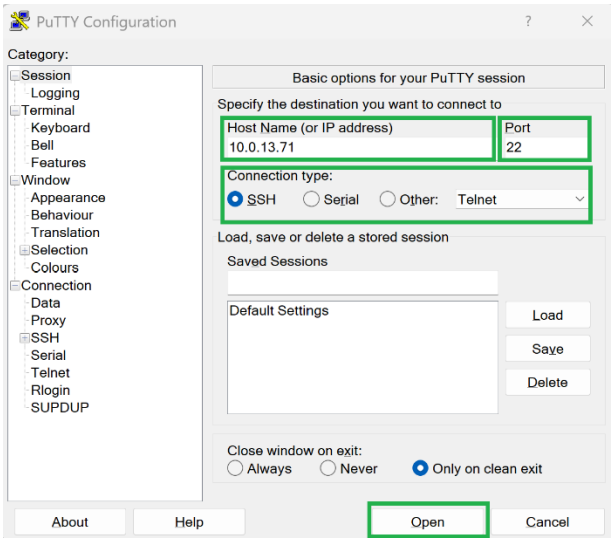
Step 3: Accessing CAT9K-CL via SSH

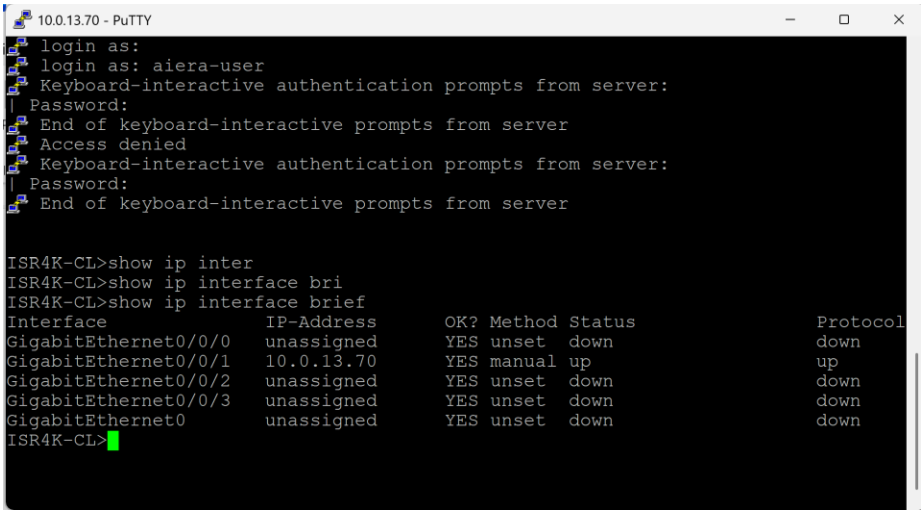
The next SEA01-103 asset that you will access is a Cisco C9300X-48HX switch. Using PuTTY, you will connect to the command-line interface via SSH. The access information is provided below.

Host Name	Mgmt IP	Access Method	Username	Password
CAT9K-CL	**10.0.13.71**	**SSH**	**aiera-user**	**Ciscolive!135**

a. Open **PuTTY**, then enter **10.0.13.71** in the *Host Name (or IP Address)* field.
b. Verify that *Port* is **22** and *Connection type:* is **SSH**, then select **Open**.



c. The asset terminal will open. At the login prompt, enter the username: **aiera-user** and press *Enter*.
d. At the password prompt, enter the password: **Ciscolive!135** and press *Enter*. When entered correctly, you will arrive at the enable prompt for *CAT9K-CL*.
e. At the enable prompt, enter the command: **show ip interface brief** and press *Enter* to simulate additional network traffic to this asset. [edit picture]



f. Close the terminal window to exit the session.

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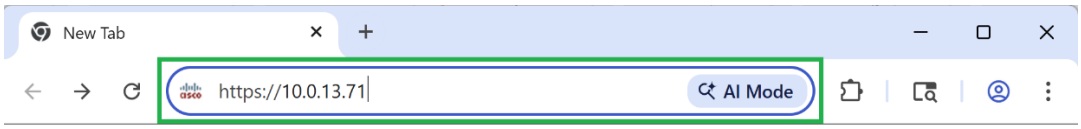
Step 4: Accessing CAT9K-CL via HTTPS

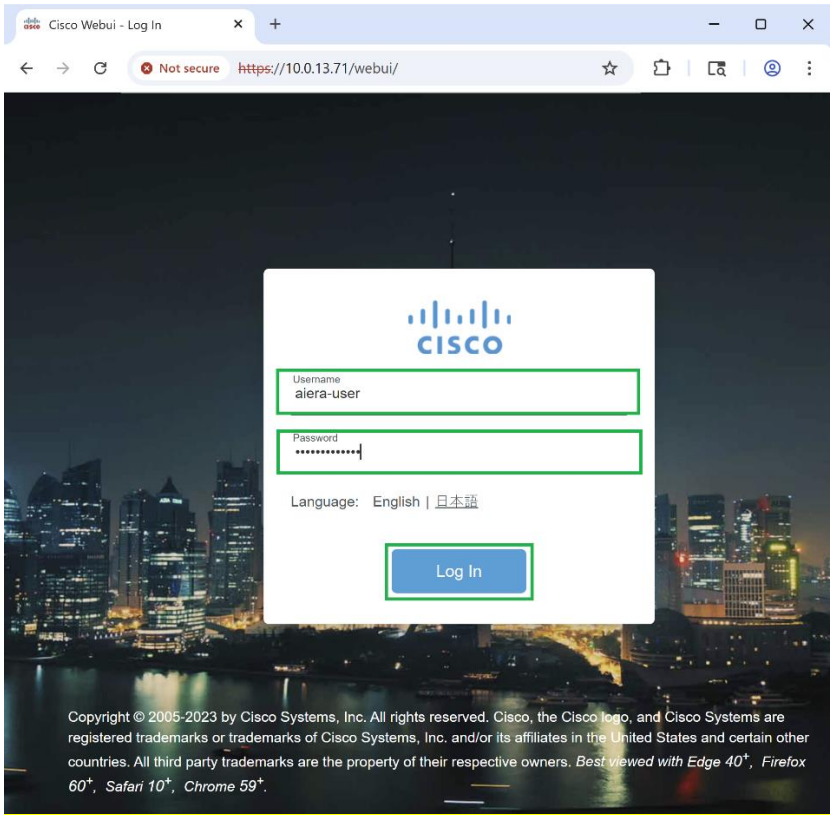
You will access *CAT9K-CL* again, but through the HTTPS protocol this time. Using varied access protocols, you will test the SNA's capability to organize flow data by fields other than IP Address. The assets access information is listed below.

Host Name	Mgmt IP	Access Method	Username	Password
CAT9K-CL	**10.0.13.71**	**HTTPS**	**aiera-user**	**Ciscolive!135**

a. Open **Google Chrome**, enter or paste the URL: **https://10.0.13.71** into the search-bar, then press *Enter*.



b. At the *CAT9K-CL*'s WebUI login screen, enter the username: **aiera-user@ciscolivevegas.com** and password: **Ciscolive!135**, then select **Log In** to access the WebUI dashboard.



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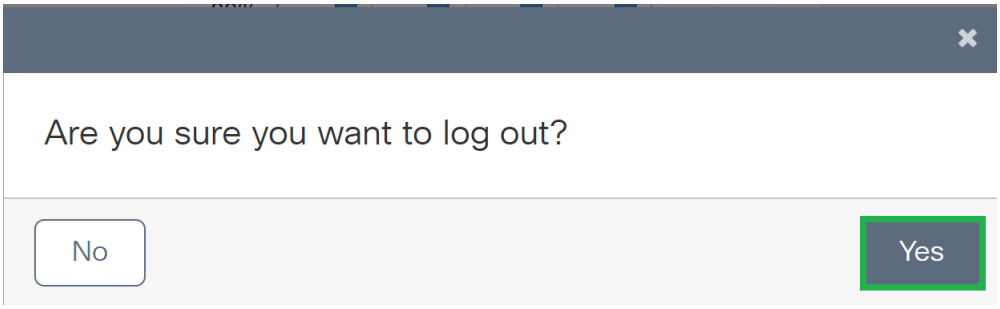
c. Feel free to navigate the WebUI to generate additional traffic associated with your laptop IP Address to monitor utilization.

d. At the page's top-right corner, select the *Log Out* button. A pop-up will appear that asks, "Are you sure you want to log out?", select *Yes*. Close the browser window after successfully logging out.



[Task 3 Summary]

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Task 4: Analyzing SEA01-103 NetFlow – Flow Search

[Task 4 Summary/Expectation/Purpose]

After simulating network traffic within your SEA01-103 Host Group, you will now analyze the recently captured Netflow on the SNA Manager. Using Flow Search, you will generate a report that provides insight into the utilization of your SEA01-103 assets.

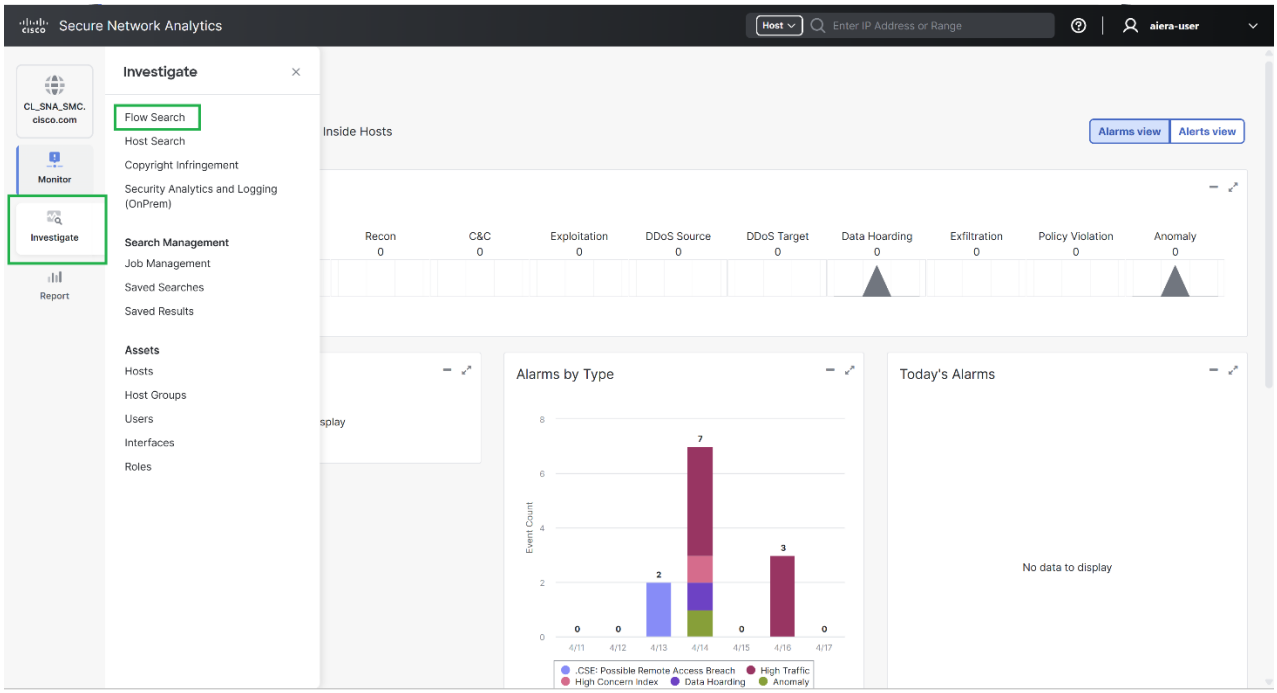
Step 1: Accessing Flow Search in SNA Dashboard

a. Open **Google Chrome**, enter or paste this URL in the search bar: https://10.0.13.50/
b. Enter your username and password, then select **Sign In**.

Username	**aiera-user**
Password	**Ciscolive!246???**

[Log In screen Pic]

c. Navigate to **Investigate** > **Flow Search**.



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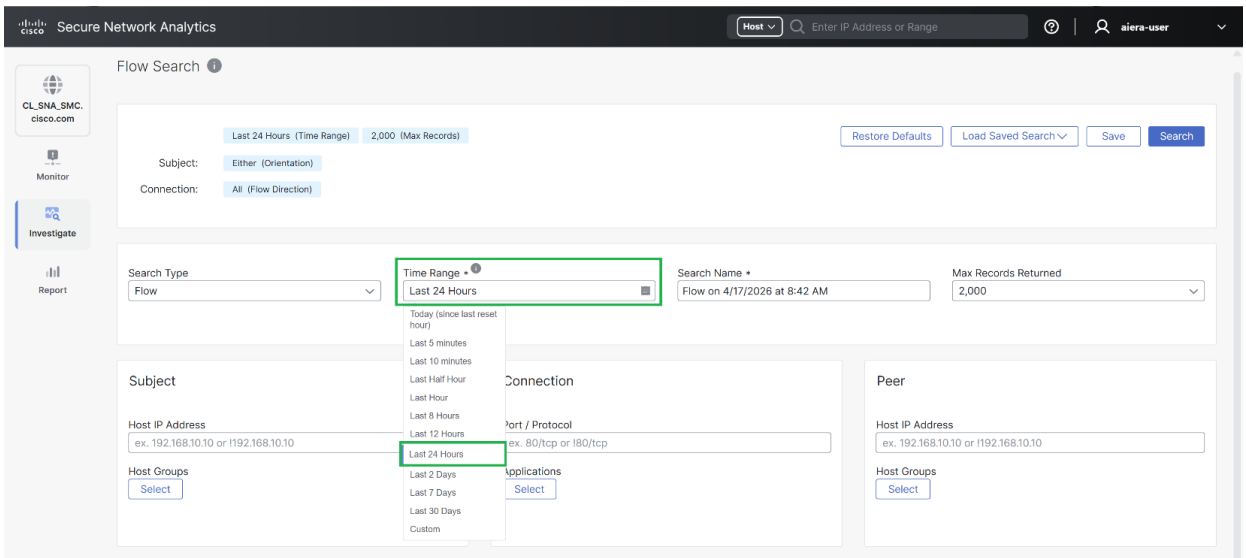
Step 2: Setting Flow Search Criteria

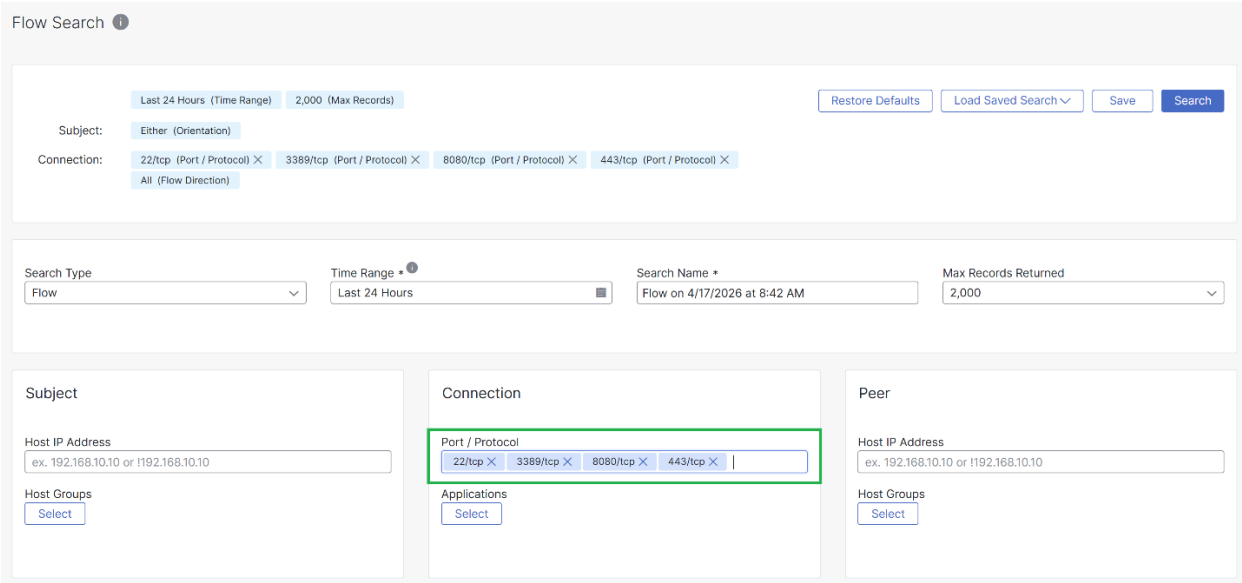
[Brief Summary/Purpose Flow Search] View laptop ip in asset utilization report

a. Click the *Time Range* dropdown and select **Last 24 hours**



b. In the *Port / Protocol* field, enter each of the protocols below. To correctly enter them, you must type each individually, followed by pressing the *Spacebar*.

- **22/TCP** > **[*Spacebar*]**
- **3389/TCP** > **[*Spacebar*]**
- **8080/TCP** > **[*Spacebar*]**
- **443/TCP** > **[*Spacebar*]**



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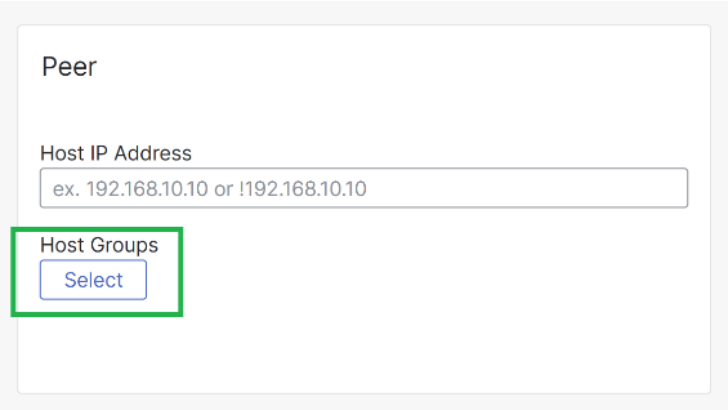
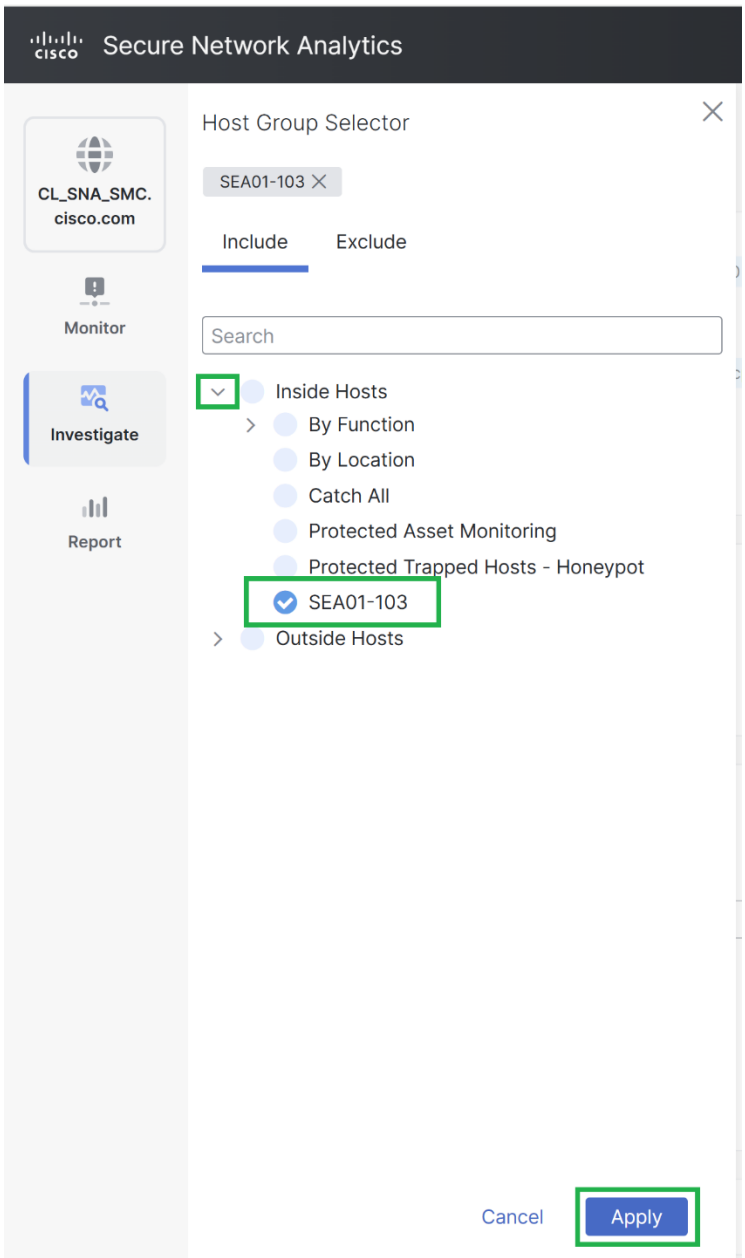
c. In the *Peer* tile, under *Host Groups*, click **Select**. [Need Full-Screen Pic]



d. In the side-panel, click **(>)** beside *Inside Hosts* to expand the dropdown, select **SEA01-103** and then **Apply**. [Need Full Screen Picture]



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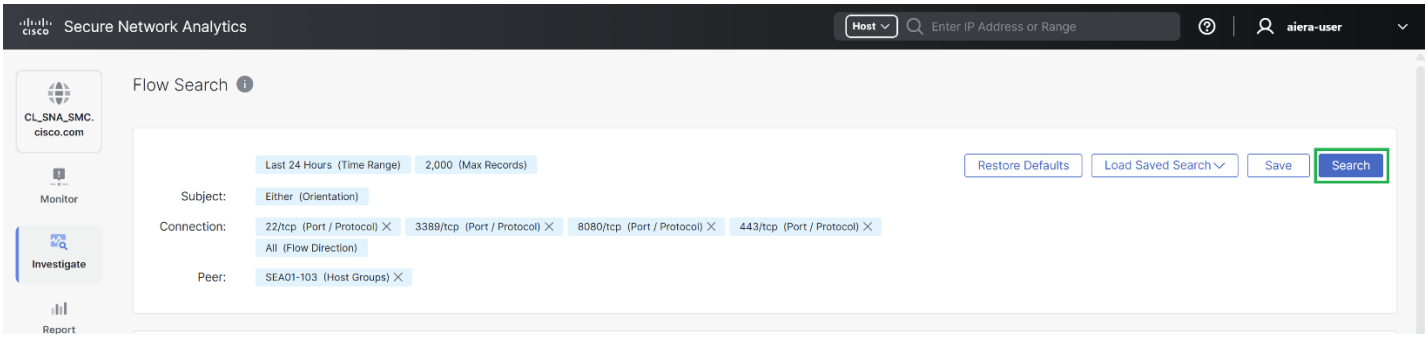
e. Now that your Flow Search criteria are set, select *Search* to generate the results.



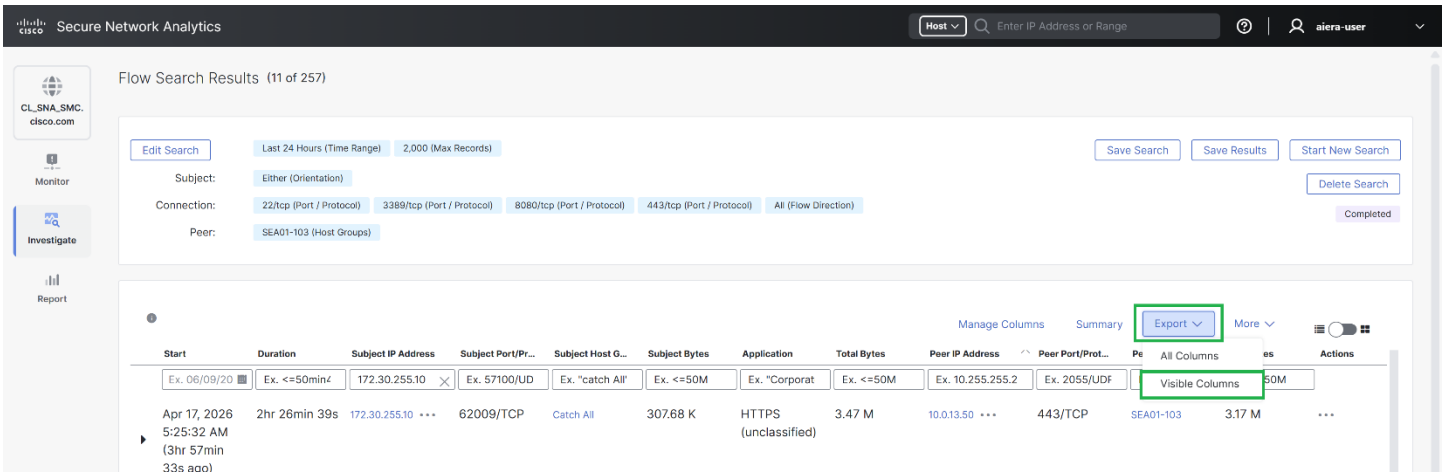
A Flow Search can produce a substantial number of results, including information that may not be relevant. For your asset utilization report, you only want flow data related to the simulations completed in Task 3. With the filters on the result table, you can enter your IP Address that was noted down in the previous task, allowing you to quickly and effectively find only the necessary data.

f. In the *Subject IP Address* filter, enter your laptop **IP Address (*Client Address IPv4*)** recorded in ***Task 3 – Step 1f***.

Note: If you do not have your IP Address, open **Cisco Secure Client** and repeat Steps 1d – 1f. In the example, our IP Address was 172.30.255.10.

Now, the table should only be populated with flows from your own connection sessions. These Flow Search results can now be exported as a csv file containing the filtered data in the table

g. Select **Export**, then **Visible Columns**, to download your asset utilization report.



h. Close the browser window to end your session.

[Task 4 Conclusion] You have successfully configured the host group and performed a flow search for the SEA01-103 to determine the device utilization. The flow data is collected by the flow collector and securely stored in the SMC Data Node. Next, we will review this data in Splunk, where our dashboard has been configured to connect directly to the Data Node database, enabling comprehensive visualization and analysis of the collected telemetry. This integration allows you to gain actionable insights into device utilization and network activity through the Splunk interface.[Rewrite]

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Task 5: Splunk dashboard view of devices monitored by Cisco Secure Network Analytics

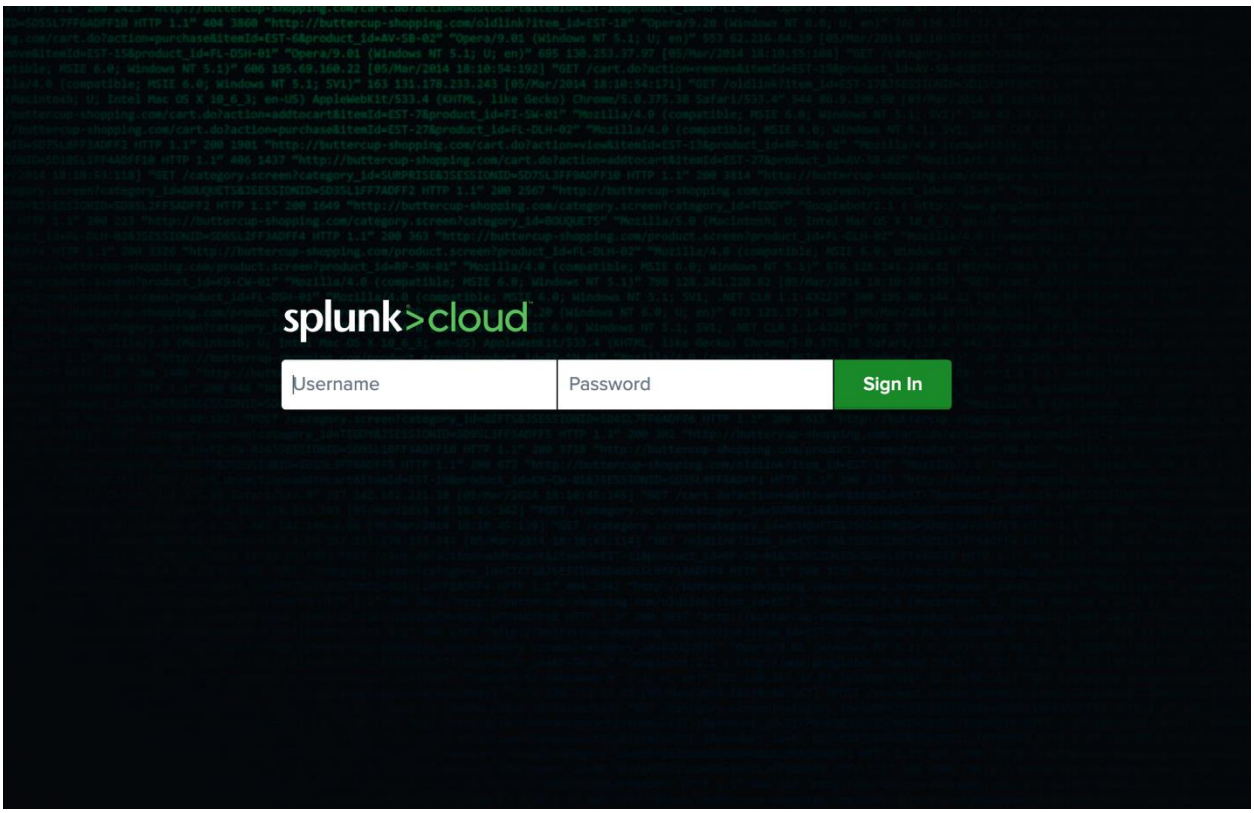
Step 1: Accessing the SNA Access Count Dashboard

To ensure the best experience and consistent performance during your lab, please use the following instructions:

Dashboard Access:

To access the Data Center Dashboard, please use Google Chrome. Chrome is the preferred browser for this lab to ensure optimal dashboard functionality and system stability.

1. Click on the link to access AI Era Power Management Dashboard on Chrome: Link
2. Logging into the dashboard by using these credentials
    a. Username:
    b. Password:



Step 2: Set the filter to SEAT01-103

Use the filter to set the Site as **Seattle** and Data Center as **SEA01-103**

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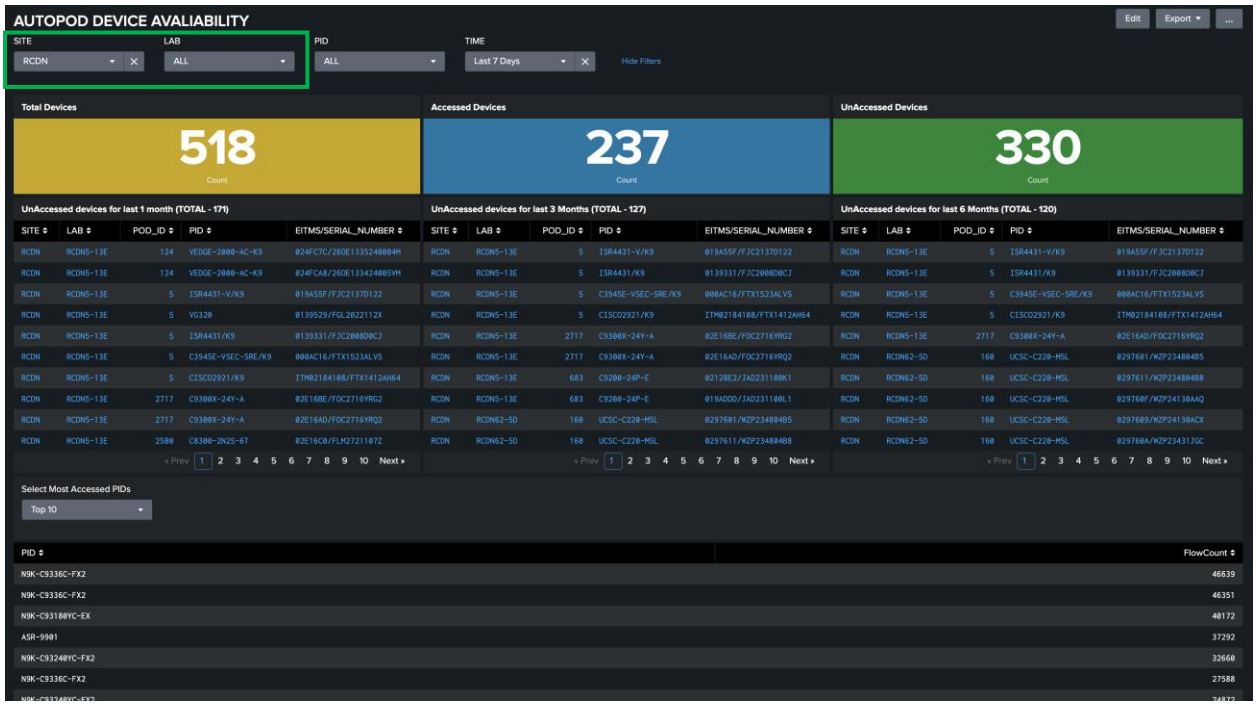

Step 2: Analyzing Access devices

This section of the interface provides real-time visibility into the data center inventory. Use the following metrics to monitor your environment:

- Total Devices: Displays the aggregate count of all devices currently provisioned or discovered within this data center. Use this to verify your baseline inventory.
- Accessed Devices: Provides a breakdown of the connection methods used for each device. Refer to this to confirm that devices are communicating through the expected protocols.

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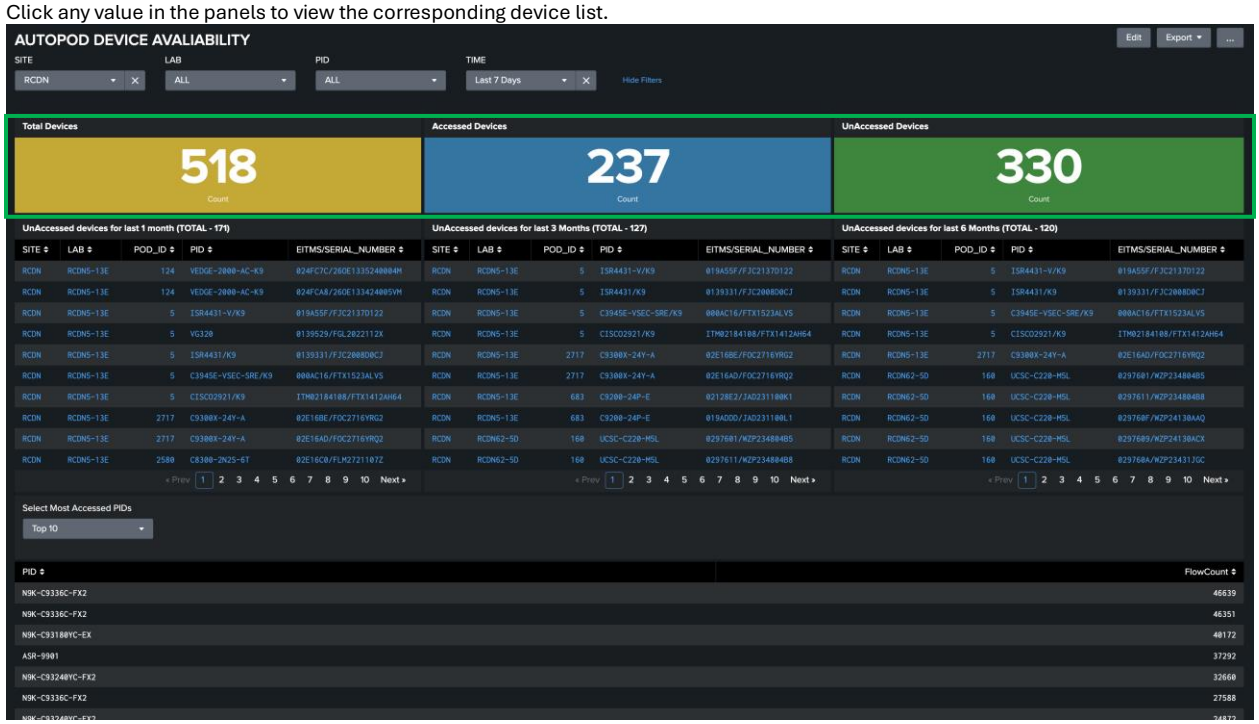
- Unaccessed Devices: Lists or quantifies devices that have not yet been reached or successfully connected to. Use this to identify potential connectivity issues or devices that require manual intervention.

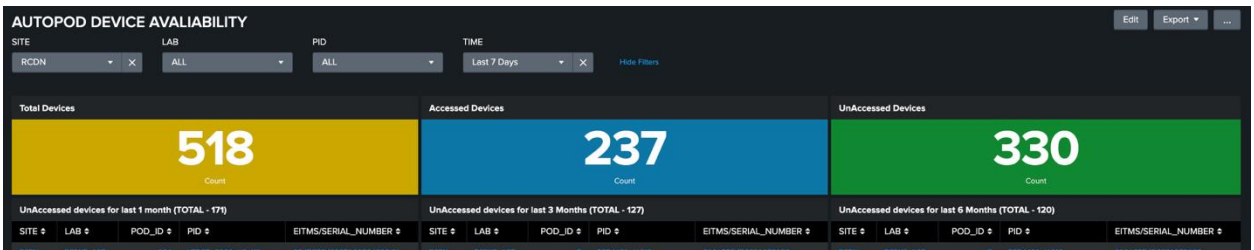
Click any value in the panels to view the corresponding device list.



Step 3: Analyzing unaccessed Devices

These three panels display a list of devices that have remained unaccessed over the past 1, 3, and 6 months. You can use this data to identify stale inventory or devices that may require decommissioning.

- 1-Month View: Identifies devices with recent connectivity issues.
- 3-Month View: Highlights devices that have been inactive for a full quarter.
- 6-Month View: Indicates long-term inactive devices that may no longer be in use.

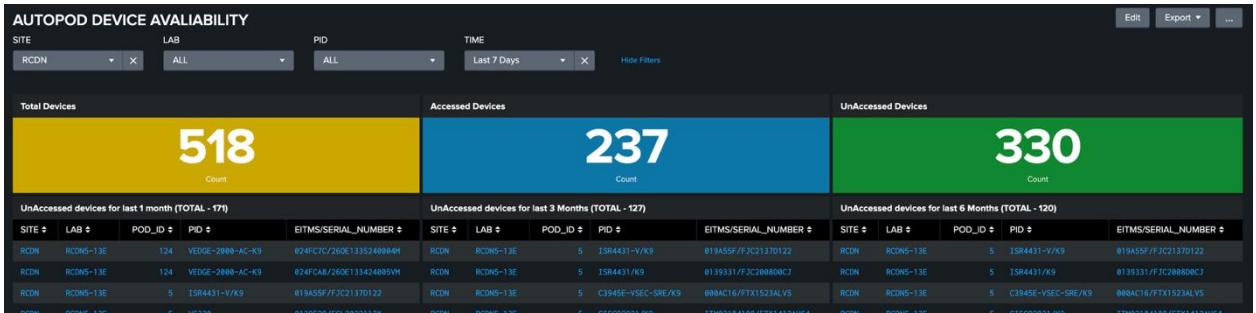


Step 4: Analyzing Top Accessed Product IDs (PIDs)

This table displays the most frequently accessed device PIDs based on the timeframe selected in the global filter.

- **PID**: The unique Product ID of the device model.
- **flowCount:** The total number of times devices with this PID were accessed during the selected period.

Interaction Tip: Use the global filter at the top of the page to adjust the reporting window. Review the **flowCount** column to determine which hardware models are the most active in your data center.



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Conclusion

Summary

Congratulations! You have successfully completed this lab on using the SNA Splunk dashboard to monitor devices within the SEATTLE data center. By analyzing access patterns and device utilization, you are now equipped to make data-driven decisions to optimize your infrastructure, such as decommissioning underutilized hardware and reallocating resources to support high-demand assets.

Thank you for attending the lab !

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